

SkillCortex

AI-Powered Job Discovery, Technical Assessment & Intelligent Interview Platform

Document Type: Product Requirements Document

Version: 1.0

Status: Product Definition / Development Blueprint

Product: SkillCortex

Primary Stack: MERN + AI

Frontend: React.js + Vite + Tailwind CSS + Framer Motion

Backend: Node.js + Express.js

Database: MongoDB + Mongoose

AI: Google Gemini API

Realtime: WebRTC + Socket.io

Authentication: JWT + Google OAuth + OTP Verification

Payments: Stripe

Deployment: Vercel + AWS + Render

1. Executive Summary

SkillCortex is an AI-powered recruitment and skill-development platform that combines:

1. Job discovery and applications
2. AI-powered resume analysis
3. Automated technical interviews
4. Real-time video/audio interviewing
5. Coding assessments with an online code editor
6. Test-case-based code evaluation
7. Interview integrity monitoring
8. AI-generated candidate scoring and feedback
9. Automated candidate-to-HR communication
10. Technical learning, skill badges and gamification
11. Premium courses and subscription/payment features

The core differentiator is:

PRD for SkillCortex

Apply for a job → upload resume → AI understands the job + resume → AI Interviewer conducts a personalized interview → coding task is evaluated → AI analyzes the entire session → candidate receives a report → qualified candidates are automatically forwarded to the HR/recruiter who posted the job.

SkillCortex should feel like a combination of:

Job Platform + AI Interviewer + Coding Assessment Platform + Learning Platform + Recruiter ATS

2. Product Vision

SkillCortex's vision is to create an intelligent hiring ecosystem where a candidate does not simply submit a resume.

Instead:

Job → Resume → AI Matching → AI Interview → Coding Assessment → Integrity Analysis → Candidate Score → HR Shortlist

The system should reduce repetitive technical screening work for HR teams while giving candidates a realistic opportunity to demonstrate their actual skills.

3. Product Goals

3.1 Candidate Goals

Candidates should be able to:

- Discover jobs
- Search and filter jobs
- View complete job descriptions
- Apply using a profile/resume
- Continue with Google
- Verify mobile number
- Verify email
- Enter an AI interview after applying

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- Complete technical questions
- Answer through voice/video
- Complete coding tasks
- Receive AI-generated feedback
- Earn Skill Points
- Unlock courses and learning content
- Earn skill badges
- Track applications and interview history

3.2 Recruiter / HR Goals

Recruiters should be able to:

- Create company profile
- Post jobs
- Define required skills
- Define interview duration
- Define interview difficulty
- Define coding requirements
- Define test cases
- Receive applications
- View AI candidate score
- Review resume
- Review transcript
- Review coding results
- Review integrity events
- Shortlist candidates
- Reject candidates
- Contact candidates
- Download candidate reports

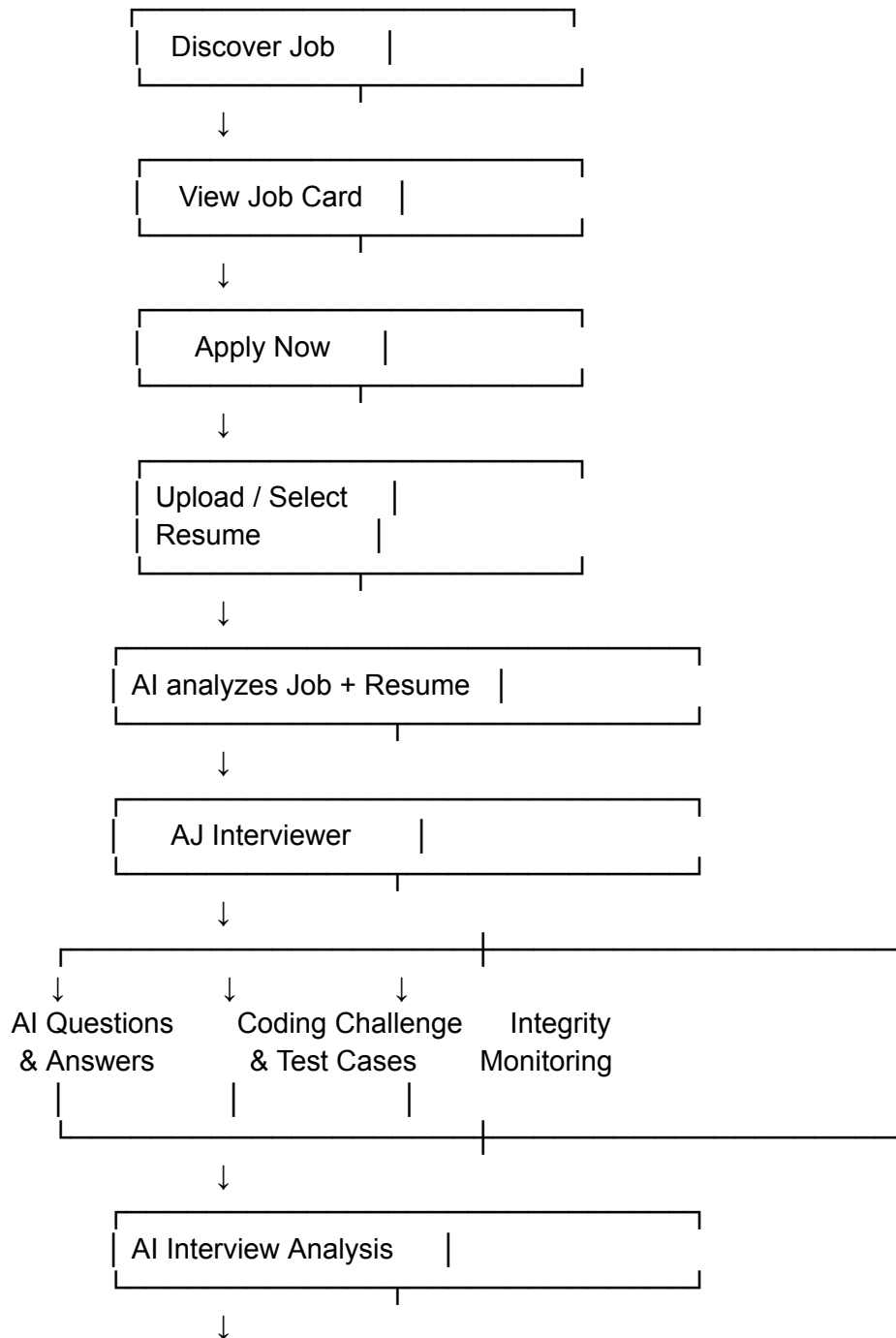
3.3 Platform Goals

SkillCortex should:

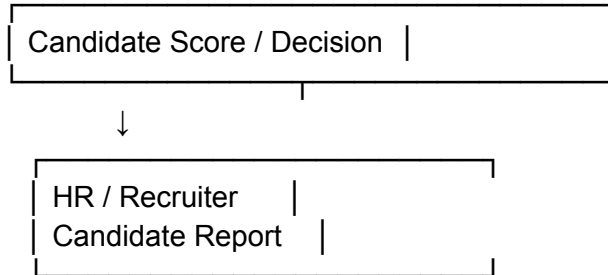
- Automate first-round technical screening
- Reduce manual HR screening
- Personalize interviews
- Evaluate technical answers
- Evaluate coding solutions
- Detect suspicious assessment behavior
- Generate actionable feedback
- Recommend learning content based on candidate weaknesses

4. Core Product Concept

The SkillCortex Loop



PRD for SkillCortex



5. Frontend Product Reference

The SkillCortex job discovery interface should take inspiration from modern job platforms such as Apna:

- Search-first interface
- Job cards
- Filters
- Salary information
- Job type
- Work mode
- Experience
- Location
- Department/category
- Sort options
- Employer information
- Clear Apply CTA

Apna currently exposes filters such as date posted, salary, work mode, work type, shift, department and sorting, making this a useful reference for SkillCortex's discovery architecture.

SkillCortex should **not copy the visual design directly**. It should create a more modern AI-focused interface.

6. Main Navigation

Candidate Navbar

SkillCortex Logo

PRD for SkillCortex

Jobs
Skill Assistant
Skill Badge
Courses
Blog
My Applications
Interviews

Coins: 1,240

 Notifications

Profile

Recommended visual language:

- White background
- Black typography
- Dark charcoal
- Light green
- Electric/modern blue accent
- Soft gray surfaces
- Minimal gradients
- Strong spacing
- Premium enterprise dashboard feel

Avoid excessive emoji usage in the final UI.

Use professional iconography through libraries such as Lucide or another consistent SVG icon system.

7. Homepage

Hero Section

Headline:

Your Skills Should Speak Before Your Resume Does.

Subheading:

PRD for SkillCortex

Discover jobs, complete AI interviews, prove your technical skills, and get evaluated automatically.

Primary CTA:

Find Jobs

Secondary CTA:

Start AI Mock Interview

Homepage Modules

- Job search
 - AI interview CTA
 - Recommended jobs
 - Trending skills
 - Skill badges
 - Latest technical articles
 - Popular courses
 - AI Assistant
 - Recruiter CTA
 - Candidate success statistics
-

8. Job Discovery Page

Route:

/jobs

Layout

Navbar

Search Jobs / Skills / Company / Location

| Filters | Job Results |
| | |

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Date Posted	Job Card	
Salary	Job Card	
Work Mode	Job Card	
Experience	Job Card	
Job Type	Job Card	
Department	Job Card	
Skills	Job Card	

Filters

- Search
 - Location
 - Salary
 - Experience
 - Fresher/Experienced
 - Work from Home
 - Hybrid
 - Office
 - Full Time
 - Part Time
 - Internship
 - Department
 - Technology
 - Posted date
 - Company
-

9. Job Card

Each job card should display:

Company Logo

Senior Full Stack Developer

Company Name

₹6 LPA – ₹10 LPA

Bengaluru / Remote

2–4 Years Experience

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React • Node.js • MongoDB • AWS

AI Assessment Available

[View Job] [Apply Now]

Additional indicators:

- Easy Apply
 - AI Interview
 - Coding Assessment
 - Verified Employer
 - Urgent Hiring
 - Remote
 - Premium Employer
-

10. Job Details Page

Route:

`/jobs/:jobId`

Sections

Job Header

- Company
- Job title
- Location
- Salary
- Experience
- Employment type
- Posted date
- Applicants
- Required skills

Job Description

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- Responsibilities
- Requirements
- Preferred qualifications
- Benefits
- Company information
- Interview process

AI Assessment Section

Example:

AI Technical Assessment

Interview Duration

30 minutes

Technical Questions

12

Coding Challenge

1

Difficulty

Intermediate

Skills Tested

React

Node.js

MongoDB

REST APIs

JavaScript

[Apply & Start Assessment]

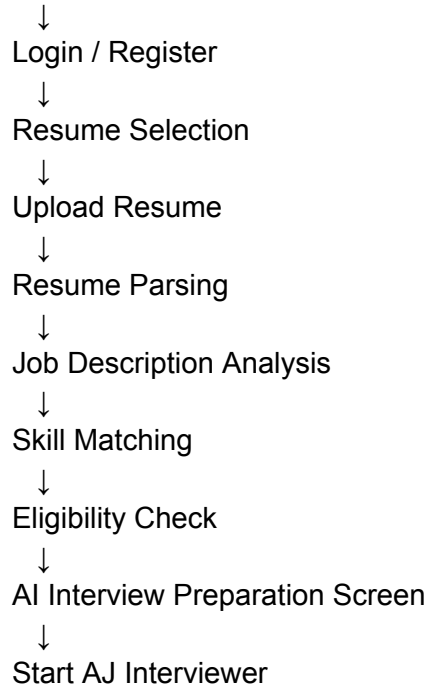
11. Apply Flow

The Apply button is the main AI transformation point.

Flow

Apply Now

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Resume formats:

- PDF
- DOCX
- TXT

The AI should extract:

- Name
- Education
- Experience
- Projects
- Skills
- Technologies
- Certifications
- Achievements
- GitHub
- LinkedIn
- Relevant keywords

12. Resume Intelligence Engine

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The system should create a normalized candidate profile.

Example:

```
{
  "skills": [
    "React",
    "Node.js",
    "MongoDB",
    "Express",
    "AWS"
  ],
  "experienceLevel": "Fresher",
  "projects": [
    "NexCall",
    "Luxtra"
  ],
  "education": "B.Tech CSE",
  "matchedSkills": [
    "React",
    "Node.js",
    "MongoDB"
  ],
  "missingSkills": [
    "Docker",
    "Redis"
  ]
}
```

The job description and resume become inputs to the AI interview engine.

13. AJ Interviewer

Product Name

AJ Interviewer

Tagline:

Your AI Technical Interviewer

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AJ should appear as a dedicated interview workspace rather than a normal chatbot.

Route:

`/interview/:sessionId`

14. AJ Interviewer UI

AJ Interviewer

Job:
Senior Full Stack Developer

Candidate:
Ajahar Husain

Time Remaining:
27:42

Candidate Video	AJ Interviewer
WEBCAM	AI Avatar / Voice
	Question

Current Question

"Explain how WebRTC establishes a peer-to-peer connection."

[Speak Answer]

Integrity
Camera ✓
Microphone ✓
Fullscreen ✓

Tab Focus ✓

[End Interview]

15. Interview Duration Engine

Interview duration must be configurable by the job creator.

Supported configurations:

- 10 minutes
- 15 minutes
- 30 minutes
- 45 minutes
- 60 minutes

The system should calculate a recommended duration from:

- Job seniority
- Number of required skills
- Number of technical questions
- Coding task complexity
- Experience level
- Recruiter configuration

Example:

Job	Recommended Duration
Internship	10–15 min
Fresher	15–30 min
Junior Developer	30 min
Mid-Level Developer	45 min
Senior Developer	60 min

Recruiters may override the recommendation.

16. Interview Question Generation

AI should dynamically generate questions based on:

Job Description

+

Resume

+

Required Skills

+

Candidate Answers

+

Difficulty

Question categories:

- Resume-based questions
- Fundamentals
- Role-specific technical questions
- Scenario questions
- Debugging
- Architecture
- Behavioral questions
- Project questions
- Coding questions

AJ should not blindly ask all predefined questions.

It should adapt.

Example:

Candidate gives a strong answer on React.

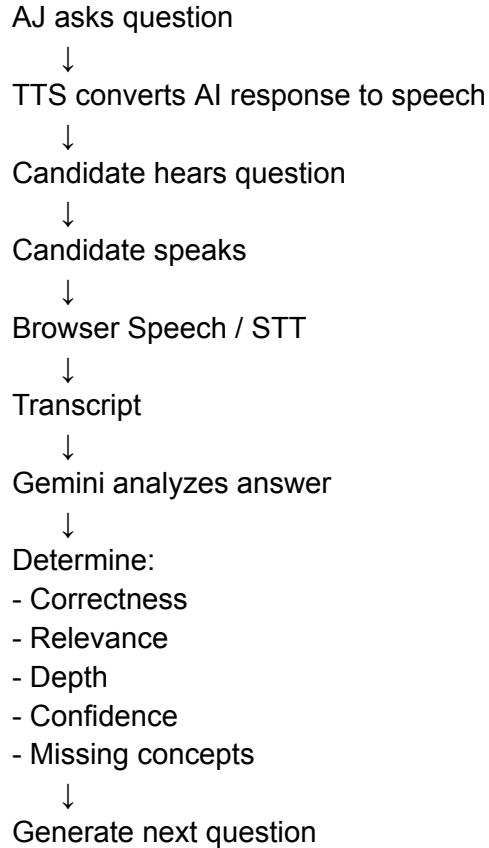
AJ increases difficulty.

Candidate struggles with Node.js.

AJ asks a simpler follow-up question.

This produces an adaptive interview.

17. Real-Time Interview Loop



18. Voice and Video

Technologies:

- WebRTC
- Browser Media APIs
- Speech-to-Text
- Text-to-Speech
- Socket.io

WebRTC should handle:

- Camera

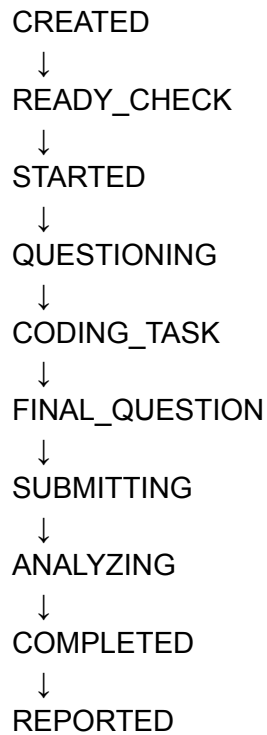
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- Microphone
- Real-time media

Socket.io should handle:

- Interview events
 - Signaling
 - Session status
 - Question synchronization
 - Evaluation events
 - Timer synchronization
 - Integrity events
-

19. Interview State Machine



Possible abnormal states:

DISCONNECTED
PAUSED
FAILED
CANCELLED

FLAGGED

20. Coding Environment

Certain jobs should include a coding challenge.

Route:

`/assessment/:assessmentId/code`

IDE Layout

Problem	Code Editor
Two Sum	Monaco Editor

Given an array...

Input Example	function solution() {
[1,2,3]	}

Expected Output

[0,1]

Test Cases

Passed: 8/10

[Run Code] [Submit Solution]

Recommended editor:

Monaco Editor

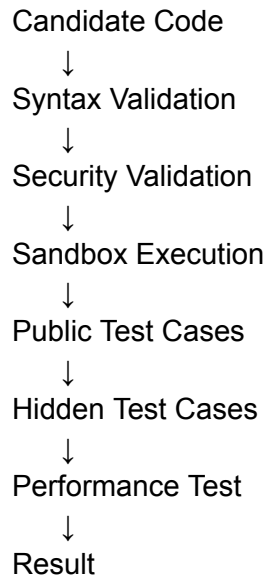
Supported languages can initially include:

- JavaScript
 - Java
 - Python
 - C++
 - SQL
-

21. Coding Evaluation Engine

The coding system should execute candidate code in an isolated sandbox.

Flow:



Evaluation metrics:

- Test cases passed
- Runtime
- Memory usage
- Compilation errors
- Runtime errors
- Correctness
- Edge-case handling

Example:

Test Cases: 10

Passed: 9

Failed: 1

Runtime: 82 ms

Memory: 41 MB

Score: 90/100

22. Anti-Cheat and Interview Integrity

This should be treated as **integrity monitoring**, not as an infallible lie detector.

The system should generate evidence and risk signals rather than automatically accusing a candidate of cheating.

Signals

Tab Switching

Detect:

- `visibilitychange`
- `blur`
- `focus`
- Fullscreen exit

Log:

Tab switched

Time: 14:23:17

Duration: 8 seconds

Camera

Check:

- Camera permission
- Camera disconnect

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- Face presence
- Multiple faces
- Prolonged absence

Attention / Eye-Gaze

Computer vision can estimate:

- Face orientation
- Gaze direction
- Long periods away from the screen

Store aggregate events rather than continuously exposing unnecessary biometric data where possible.

Example:

Attention Signal

Normal: 91%

Away from screen: 9%

Multiple Person Detection

If more than one person is visible:

Integrity Warning

Additional person detected in frame.

Please ensure you are alone during the interview.

Noise Detection

Use microphone audio analysis to identify:

- Multiple voices
- Significant background speech
- Strong environmental noise
- Audio interruptions

The system should label this as:

Background Audio Signal

rather than automatically deciding that another person assisted the candidate.

23. Browser / Extension Control

A normal web page cannot reliably and universally "block all browser extensions."

Therefore the requirement should be implemented as:

Secure Assessment Mode

During an interview:

- Force fullscreen
- Detect fullscreen exit
- Detect tab switch
- Detect window blur/focus
- Disable unnecessary navigation
- Disable copy/paste where appropriate
- Disable right-click where appropriate
- Warn against developer tools
- Detect suspicious browser behavior where technically possible
- Use secure browser/assessment environment for higher-security recruiter assessments

For high-stakes hiring, the platform should offer an optional controlled assessment environment rather than claiming that arbitrary browser extensions can be completely disabled.

24. Integrity Score

Example:

Technical Score	84
Coding Score	92
Communication	79
Problem Solving	88
Integrity Signals	94
Overall	87

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Integrity score should be accompanied by evidence.

Example:

Integrity Events

Tab switches: 2

Fullscreen exits: 1

Multiple-person event: 0

Audio anomaly: 1

Camera interruption: 0

HR should be able to inspect events before making decisions.

25. AI Answer Evaluation

Every answer should receive multiple dimensions.

Example:

Question:

What is the difference between let, var and const?

Evaluation

Correctness: 4.5/5

Relevance: 5/5

Technical Depth: 4/5

Clarity: 4/5

Confidence: 4/5

Score: 4.4/5

AI should assess:

- Accuracy
- Relevance
- Completeness
- Depth
- Reasoning

- Practical understanding
 - Communication
 - Consistency
-

26. Anti-Hallucination Requirement

AI must not award a high score merely because an answer sounds professional.

Evaluation should be grounded in:

- Expected concepts
- Reference answer
- Job requirements
- Technical correctness
- Coding result
- Follow-up consistency

For important questions, the AI evaluator should produce structured evidence:

```
{
  "score": 4.2,
  "correctConcepts": [
    "closure",
    "lexical scope"
  ],
  "missingConcepts": [
    "execution context"
  ],
  "errors": [
    "Incorrect claim regarding memory allocation"
  ]
}
```

27. Final Interview Analysis

After completion:

Interview Completed ✓

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Analyzing your performance...

Resume Match
Technical Knowledge
Coding
Communication
Problem Solving
Integrity Signals

Generating final report...

Then:

Interview Report

Overall Score: 86/100

Resume Match: 91%
Technical: 84%
Coding: 90%
Communication: 79%
Problem Solving: 87%

Recommendation:
Strong Technical Candidate

28. Candidate Result Page

Route:

`/interview/:id/result`

Sections:

Overall Score

Large visual score.

Strengths

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Example:

- Strong JavaScript fundamentals
- Good REST API knowledge
- Strong React experience

Weaknesses

Example:

- Weak system design
- Limited caching knowledge
- Needs stronger SQL optimization

Question Analysis

Every question:

Question 01

Score 4.5/5

Your answer

...

AI Review

...

What you should improve

...

Coding Analysis

9/10 Test Cases Passed

Time Complexity:

$O(n)$

Suggested:

$O(n)$

Code Quality:

8.5/10

29. HR Notification System

After interview analysis, the recruiter who posted the job should receive the candidate result.

Example email:

Subject:

New SkillCortex Technical Assessment Result — Senior Full Stack Developer

Candidate:

Ajahar Husain

Overall Score:

86/100

Technical:

84/100

Coding:

90/100

Recommendation:

Strong Candidate

[View Candidate Report](#)

Email notifications can also be sent when:

- Candidate applies
- Candidate starts assessment
- Candidate completes assessment
- Candidate is shortlisted
- Candidate is rejected
- Recruiter changes status

30. HR Dashboard

Route:

</hr/dashboard>

Main Navigation

Dashboard
Jobs
Candidates
Assessments
Interview Reports
Analytics
Messages
Company
Billing
Settings

Dashboard Cards

Active Jobs	12
Applications	428
Interviews	164
Qualified	52
Shortlisted	31

31. Job Creation

HR should be able to configure:

Job Title
Company
Location
Salary
Experience
Description
Responsibilities
Requirements
Skills
Interview Duration
Interview Difficulty
Coding Assessment
Number of Questions
Passing Score

AI Interview Configuration

AI Interview: ON

Duration:
30 minutes

Difficulty:
Intermediate

Coding:
Required

Minimum Score:
70%

Integrity Monitoring:
ON

AI Recommendation:
ON

32. Candidate Ranking

HR should receive a ranking table.

Candidate	Score	Coding	Technical	Status
Candidate A	91	94	89	Strong
Candidate B	86	90	84	Strong
Candidate C	73	62	78	Review
Candidate D	51	44	57	Reject

Filters:

- Score
- Skill
- Experience
- Coding score
- Location

- Interview completed
 - Integrity status
 - Shortlisted
 - Rejected
-

33. Skill Assistant

Navbar:

Skill Assistant

Route:

`/assistant`

This is the candidate's AI career assistant.

Capabilities:

- Explain technical concepts
- Mock interview practice
- Resume suggestions
- Job preparation
- Coding help
- Learning roadmap
- Interview feedback
- Skill-gap analysis

Example:

You:

I have a React interview tomorrow.

Skill Assistant:

Based on your profile, your strongest areas are React Hooks and API integration.

You should revise:

1. useEffect
2. useMemo
3. useCallback
4. Context API

5. React rendering

34. Skill Badge System

Route:

`/skills`

Candidates can earn badges such as:

React Developer
Node.js Developer
Java Backend
MongoDB
AWS Cloud
Full Stack
Problem Solver
AI Fundamentals

Badge levels:

Bronze
Silver
Gold
Platinum
Elite

Badges are earned from:

- Interviews
 - Coding assessments
 - Quizzes
 - Courses
 - Projects
 - Verified recruiter assessments
-

35. Coin / Skill Point Economy

Name:

Skill Coins

Candidates earn coins by:

- Completing interviews
- Passing quizzes
- Completing courses
- Solving coding challenges
- Earning badges
- Maintaining learning streaks

Example:

AI Interview Completed	+100
Quiz Passed	+50
Coding Challenge	+75
Course Completed	+200
Badge Earned	+150

Coins can unlock:

- Premium notes
- Premium lectures
- Advanced challenges
- Mock interviews
- Practice packs
- AI features

The exact monetization rules should be designed so that basic job application and essential assessment access do not become pay-to-win.

36. Courses

Route:

</courses>

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Each course contains:

Course Name

Introduction

Notes

Video Lectures

Practice

Quizzes

Coding Challenges

Final Assessment

Certificate

Example:

Full Stack MERN Mastery

HTML/CSS

JavaScript

React

Node.js

Express

MongoDB

Authentication

AWS

Deployment

System Design

37. Blog

Route:

`/blog`

Content should be AI-assisted but human-reviewed.

Categories:

- JavaScript
- React
- Node.js

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- MongoDB
- Java
- Python
- AWS
- AI
- Interview Preparation
- System Design
- DSA
- Career

Dynamic recommendation:

Candidate skill gap:

Node.js = Weak

Blog recommendations:

- Express middleware explained
 - Node.js event loop
 - REST API design
 - JWT authentication
-

38. Authentication

SkillCortex should support:

Standard

- Email/password
- Mobile/password or OTP

Google

Continue with Google

Google Identity Services should be used for modern Google authentication; Google's current guidance supports Sign in with Google and One Tap, with FedCM considerations for supported browser environments.

Verification

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- Email OTP
- Mobile OTP

For the MVP, use an open-source/self-hosted OTP verification approach where practical, while keeping the provider interface abstract enough to switch to a managed SMS/email provider later.

Authentication flow:

Register
↓
Email Verification
↓
Mobile Verification
↓
Profile Creation
↓
Dashboard

Google users:

Continue with Google
↓
Google Identity
↓
Backend verifies credential
↓
Create / Login User
↓
JWT Session

39. Authentication Security

Backend should implement:

- Password hashing with bcrypt
- JWT access tokens
- Refresh token strategy
- Secure HTTP-only cookies where appropriate
- Rate limiting
- OTP expiration

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- OTP attempt limits
- Account lockout/rate control
- Email verification
- Mobile verification
- Password reset
- OAuth identity linking
- Role-based authorization

Roles:

CANDIDATE

HR

ADMIN

40. Database Architecture

MongoDB collections:

users
profiles
resumes
jobs
applications
interviews
interviewQuestions
answers
codingAssessments
codingSubmissions
testCases
integrityEvents
aiEvaluations
skillBadges
skillCoins
courses
lessons
quizzes
quizAttempts
blogs
notifications
payments
subscriptions

companies
auditLogs

41. Core User Schema

```
User {  
  name,  
  email,  
  mobile,  
  passwordHash,  
  googleId,  
  role,  
  emailVerified,  
  mobileVerified,  
  profile,  
  skills,  
  coins,  
  badges,  
  createdAt,  
  updatedAt  
}
```

42. Job Schema

```
Job {  
  companyId,  
  title,  
  description,  
  responsibilities,  
  requirements,  
  location,  
  salary,  
  experienceMin,  
  experienceMax,  
  employmentType,  
  workMode,  
  skills,  
  interviewConfig,  
}
```

```
codingConfig,  
status,  
createdAt,  
updatedAt  
}
```

43. Application Schema

```
Application {  
  jobId,  
  candidateId,  
  resumeId,  
  status,  
  matchScore,  
  interviewId,  
  appliedAt,  
  updatedAt  
}
```

Status:

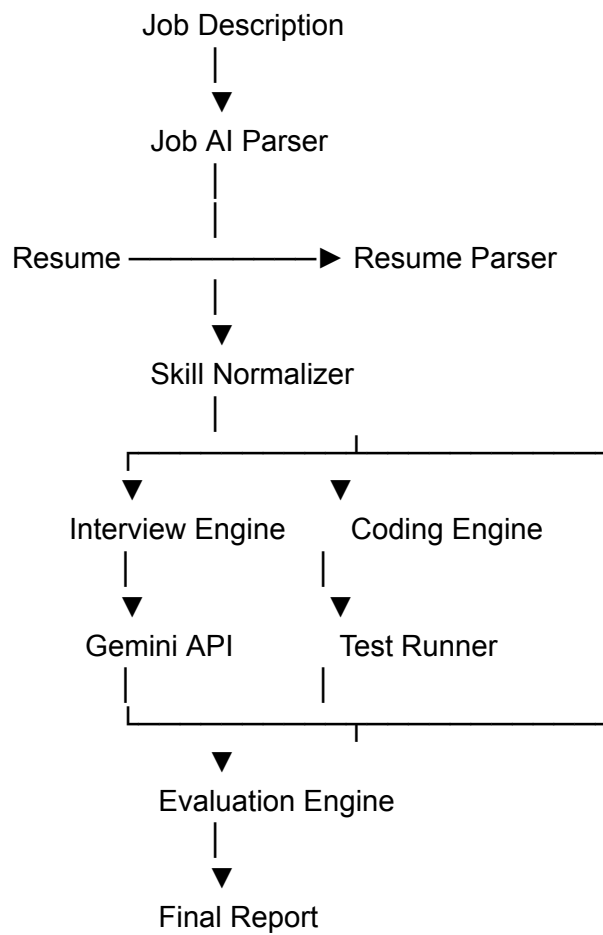
```
APPLIED  
ASSESSMENT_PENDING  
ASSESSMENT_STARTED  
ASSESSMENT_COMPLETED  
UNDER_REVIEW  
SHORTLISTED  
REJECTED  
HIRED
```

44. Interview Schema

```
Interview {  
  jobId,  
  candidateId,  
  applicationId,  
  duration,
```

```
startedAt,  
completedAt,  
questions,  
answers,  
codingAssessment,  
integrityEvents,  
transcript,  
overallScore,  
recommendation,  
status  
}
```

45. AI Evaluation Architecture



46. Gemini AI Responsibilities

Gemini should be used for:

- Resume parsing
- Skill extraction
- Job parsing
- Interview question generation
- Adaptive follow-up questions
- Answer evaluation
- Transcript analysis
- Coding explanation
- Candidate feedback
- Job-candidate matching
- Learning recommendations
- Blog assistance
- Skill-gap recommendations

Critical principle:

AI generates and evaluates; deterministic backend services enforce business rules.

Do not let the LLM decide security-critical events, payment state, authorization, or test-case execution.

47. API Architecture

Authentication

POST /api/auth/register

POST /api/auth/login

POST /api/auth/google

POST /api/auth/send-email-otp

POST /api/auth/verify-email-otp

POST /api/auth/send-mobile-otp

POST /api/auth/verify-mobile-otp

POST /api/auth/forgot-password

POST /api/auth/reset-password

POST /api/auth/refresh

POST /api/auth/logout

Jobs

GET /api/jobs
GET /api/jobs/:id
POST /api/jobs
PATCH /api/jobs/:id
DELETE /api/jobs/:id

Applications

POST /api/applications
GET /api/applications
GET /api/applications/:id
PATCH /api/applications/:id/status

Resume

POST /api/resumes/upload
GET /api/resumes
GET /api/resumes/:id

Interview

POST /api/interviews/start
GET /api/interviews/:id
POST /api/interviews/:id/answer
POST /api/interviews/:id/event
POST /api/interviews/:id/complete
GET /api/interviews/:id/report

Coding

GET /api/assessments/:id
POST /api/assessments/:id/run
POST /api/assessments/:id/submit
GET /api/submissions/:id

AI

POST /api/ai/resume-analyze
POST /api/ai/job-analyze
POST /api/ai/interview/question
POST /api/ai/interview/evaluate
POST /api/ai/feedback

48. File Storage

Resume files and other user uploads should not be stored directly inside MongoDB.

Recommended architecture:

Browser
↓
Backend / Signed Upload
↓
AWS S3
↓
S3 Object URL
↓
MongoDB stores metadata

Files:

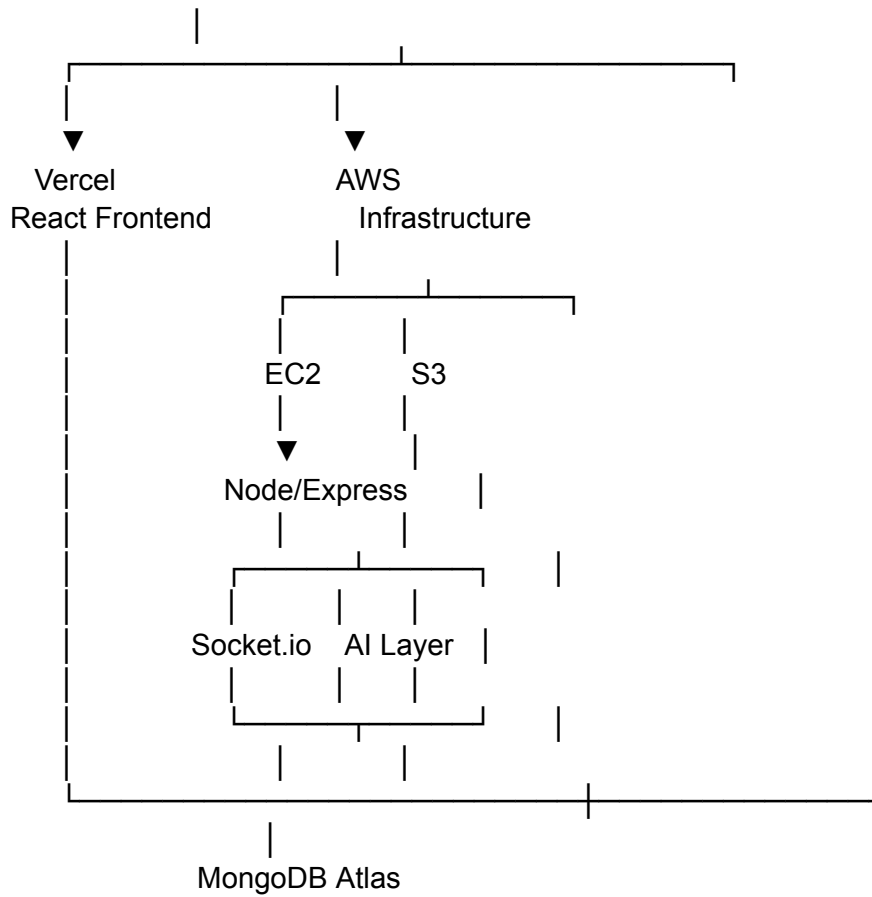
- Resume
 - Candidate documents
 - Company logos
 - Course assets
 - Optional interview recordings where consent and policy permit
-

49. Cloud Architecture

Production Architecture

INTERNET

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50. Deployment Strategy

Frontend

Vercel

Use for:

- React/Vite frontend
- CDN
- Automatic deployment
- Preview deployments
- Custom domain
- HTTPS

Backend

Primary production option:

AWS EC2

Use for:

- Node.js
- Express
- Socket.io
- WebRTC signaling
- AI orchestration

Process management:

PM2

Optional Secondary Backend

Render

Render can be used for:

- Development/staging backend
- Alternative API deployment
- Background services
- Demo deployments

Production should avoid unnecessary duplication unless there is a clear reliability reason.

51. AWS Services

Recommended AWS services:

EC2
S3
CloudFront
Route 53
IAM
CloudWatch

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Secrets Manager

SES

SNS

Potential usage:

EC2

- Node.js backend
- Socket.io

S3

- Resume/object storage

CloudFront

- CDN for static/file delivery

Route 53

- DNS

IAM

- Least-privilege permissions

CloudWatch

- Logs and monitoring

Secrets Manager

- API keys and secrets

SES

- Email notifications

SNS

- SMS/notification integration if selected
-

52. Payment System

SkillCortex Premium should use:

Stripe

Use cases:

- Premium subscription
- Course purchase
- Premium interview packages
- Advanced assessments
- Premium learning content

Suggested products:

SkillCortex Free

₹0

SkillCortex Plus

Monthly subscription

SkillCortex Pro

Monthly subscription

Premium Course

One-time purchase

Payment architecture:

Frontend



Backend creates Checkout Session



Stripe Checkout



Payment



Stripe Webhook



Backend verifies event



MongoDB subscription update

Never trust the browser alone to mark a payment as successful.

53. Premium Features

Potential premium features:

- Advanced AI mock interviews
- Unlimited practice interviews
- Advanced coding assessments
- Premium courses
- Interview analytics
- Advanced reports
- Skill-gap roadmap
- Resume optimization
- Personalized interview roadmap
- Premium masterclasses

Essential job applications should remain usable without requiring payment unless a specific employer assessment is explicitly designated as paid.

54. Notifications

Channels:

- In-app
- Email
- Optional SMS

Notification examples:

Application submitted

Interview invitation

Interview completed

Result ready

Candidate shortlisted

New recruiter message

Course unlocked
Badge earned
Coins received
Subscription renewed

55. Candidate Dashboard

Route:

[/dashboard](#)

Main widgets:

Welcome back, Candidate

Application Status

Interview Score

Skill Level

Coins

Badges

Learning Streak

Recommended Jobs

Upcoming Interviews

Weak Skills

Recommended Courses

Recent Activity

56. Job Recommendation Engine

The recommendation engine should combine:

Candidate Skills

+

Resume

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+
Experience
+
Location
+
Job Skills
+
Past Applications
+
Interview Performance
+
Learning History

Example:

92% Match
Senior React Developer

Skills Match: 94%
Experience Match: 86%
Location Match: 100%
Resume Match: 91%

57. AI Personalization

After an interview:

Weak:
System Design

Medium:
MongoDB

Strong:
React
Node.js
JavaScript

SkillCortex automatically recommends:

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Course:
System Design Fundamentals

Blog:
REST API Scalability

Practice:
10 System Design Questions

Mock Interview:
Backend Architecture

This creates a closed learning loop.

58. Admin Panel

Route:

`/admin`

Admin capabilities:

- Manage users
 - Manage recruiters
 - Manage jobs
 - Manage reports
 - Manage courses
 - Manage blogs
 - Manage badges
 - Manage subscriptions
 - View platform analytics
 - Review abuse
 - Review AI failures
 - Manage flagged interviews
 - Manage feature configuration
-

59. Security Requirements

Mandatory:

- HTTPS
 - JWT/session security
 - Password hashing
 - Input validation
 - MongoDB injection protection
 - XSS protection
 - CSRF protection where applicable
 - CORS configuration
 - Helmet
 - Rate limiting
 - API authorization
 - Role-based access
 - Secure file uploads
 - File type validation
 - Malware scanning strategy
 - Secrets management
 - Audit logging
-

60. Privacy Requirements

Because SkillCortex processes resumes, voice, video and assessment data, privacy must be a first-class requirement.

Candidate consent should be obtained before:

- Camera access
- Microphone access
- Interview recording, where enabled
- Integrity monitoring
- AI analysis

Candidate should be informed:

- What is collected
- Why it is collected
- How long it is retained
- Who may access it
- Whether it is shared with the employer

Biometric/attention signals should be handled conservatively and should not be represented as definitive proof of dishonesty.

61. AI Fairness Requirement

The platform must not automatically reject candidates solely because of:

- Accent
- Voice pitch
- Speaking speed
- Camera quality
- Skin tone
- Facial appearance
- Eye shape
- Disability-related behavior
- Temporary connectivity issues

Technical evaluation should emphasize:

What the candidate knows and demonstrates, not superficial appearance characteristics.

Human review should remain available for disputed or high-impact decisions.

62. Interview Reliability

Before starting:

Camera ✓
Microphone ✓
Internet ✓
Browser ✓
Fullscreen ✓
Audio ✓

Connectivity problems should not automatically count as negative technical performance.

The system should support:

- Reconnection
 - Session recovery
 - Server-side timer
 - Autosave
 - Answer checkpointing
-

63. Interview Timer

Timer must be controlled server-side.

Example:

Interview Duration:

30:00

Remaining:

21:43

The client only displays the timer.

The backend maintains authoritative timestamps.

This prevents candidates from manipulating the browser timer.

64. Question Adaptation Algorithm

Example:

Question 1 → Medium

↓

Score 4.5/5

↓

Increase difficulty

Question 2 → Hard

↓

Score 2/5

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↓
Follow-up clarification

Question 3 → Medium

↓
Score 3.8/5

↓
Continue

The interview should feel conversational rather than like a static questionnaire.

65. Interview Scoring Model

Example:

Technical Knowledge	30%
Problem Solving	20%
Coding	25%
Communication	10%
Job Skill Match	10%
Integrity Signals	5%

Recruiters can configure weights.

Important:

Integrity signals should remain a supporting assessment dimension rather than becoming a black-box automatic rejection mechanism.

66. Final Candidate Recommendation

Possible outcomes:

EXCEPTIONAL
STRONG
GOOD

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REVIEW
WEAK
NOT_RECOMMENDED

Example:

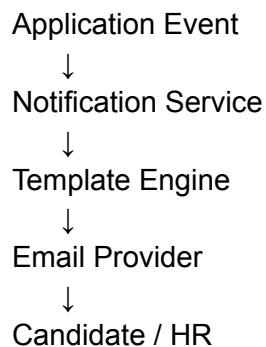
Overall:
STRONG

Confidence:
86%

Reason:
Candidate demonstrates strong React and Node.js knowledge, passed 9/10 coding tests, and showed good problem-solving ability.

Recommended HR Action:
Move to next interview round.

67. Email Architecture



Template types:

- Welcome
- OTP
- Verification
- Application confirmation
- Interview invitation
- Interview result

- HR candidate alert
 - Shortlist
 - Rejection
 - Payment receipt
 - Course purchase
 - Subscription renewal
-

68. Recommended Frontend Routes

/

- /jobs
- /jobs/:id
- /login
- /register
- /verify
- /resume
- /dashboard
- /applications
- /applications/:id
- /interviews
- /interview/:id
- /interview/:id/result
- /assessment/:id/code
- /assistant
- /skills
- /courses
- /courses/:id
- /blog
- /blog/:slug
- /profile
- /settings

/hr

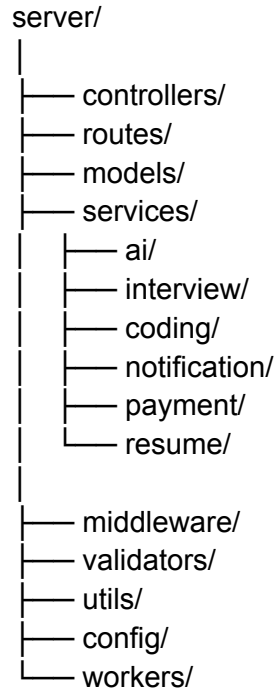
- /hr/dashboard
- /hr/jobs
- /hr/jobs/create
- /hr/jobs/:id
- /hr/candidates
- /hr/candidates/:id
- /hr/analytics
- /hr/settings

/admin
/admin/dashboard
/admin/users
/admin/jobs
/admin/courses
/admin/blogs
/admin/payments
/admin/settings

69. Frontend Component Architecture

```
src/  
├── components/  
│   ├── navbar/  
│   ├── jobs/  
│   ├── interview/  
│   ├── coding/  
│   ├── dashboard/  
│   ├── badges/  
│   ├── courses/  
│   ├── blog/  
│   └── ui/  
├── pages/  
│   ├── Home/  
│   ├── Jobs/  
│   ├── Interview/  
│   ├── Dashboard/  
│   ├── Assistant/  
│   ├── Courses/  
│   └── HR/  
├── hooks/  
├── services/  
├── store/  
├── utils/  
├── layouts/  
└── routes/
```

70. Backend Architecture



71. Service Separation

Important backend modules:

Auth Service

Authentication and authorization.

Resume Service

Upload and extraction.

Job Service

Job management.

Application Service

Candidate application lifecycle.

Interview Service

Interview session management.

AI Service

Gemini orchestration.

Coding Service

Code execution and test cases.

Integrity Service

Interview environment events.

Notification Service

Email/in-app communication.

Payment Service

Stripe transactions and subscriptions.

Learning Service

Courses, quizzes, badges, coins.

72. Coding Sandbox Security

Never execute candidate code directly on the main Node.js application server.

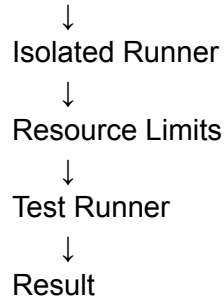
Use an isolated execution environment such as:

Submission API



Job Queue

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Apply:

- CPU limit
 - Memory limit
 - Execution timeout
 - Restricted network
 - Temporary filesystem
 - Container isolation
 - Process restrictions
-

73. Performance Requirements

Target:

- Job page load under 2–3 seconds under normal conditions
- API responses under 500 ms for common non-AI operations
- AI operations show clear loading state
- Interview events synchronized in near real time
- Coding execution result returned promptly for normal submissions

The system should use:

- Pagination
 - Lazy loading
 - CDN
 - Image optimization
 - API caching
 - Query indexes
 - Background jobs
-

74. Analytics

Candidate analytics:

- Applications
- Interview completion
- Score history
- Skill progression
- Weak skills
- Courses completed
- Coding performance

HR analytics:

- Applications per job
- Interview completion rate
- Average candidate score
- Time-to-shortlist
- Skill distribution
- Coding pass rate
- Candidate funnel

Platform analytics:

- Active candidates
 - Active recruiters
 - Assessments completed
 - AI usage
 - Premium conversions
 - Course completion
 - Revenue
-

75. Observability

Use:

- CloudWatch
- Application logs
- Error monitoring
- API latency monitoring
- WebSocket monitoring

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- AI request tracking
- Payment webhook logs
- Assessment execution logs

Each interview should have a unique session ID for debugging.

76. CI/CD

Recommended pipeline:

GitHub
↓
Pull Request
↓
Lint
↓
Unit Tests
↓
Build
↓
Security Checks
↓
Deploy

Frontend:

GitHub → Vercel

Backend:

GitHub → AWS EC2

Staging can optionally use:

GitHub → Render

77. Environment Variables

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Example:

```
MONGO_URI=  
JWT_SECRET=  
GOOGLE_CLIENT_ID=  
GOOGLE_CLIENT_SECRET=  
GEMINI_API_KEY=  
STRIPE_SECRET_KEY=  
STRIPE_WEBHOOK_SECRET=  
AWS_ACCESS_KEY_ID=  
AWS_SECRET_ACCESS_KEY=  
AWS_REGION=  
AWS_S3_BUCKET=  
SMTP_HOST=  
SMTP_USER=  
SMTP_PASSWORD=
```

Secrets must never be committed to GitHub.

78. MVP — Phase 1

The first development milestone should focus on proving the core value.

MVP Scope

Candidate

- Register/login
- Google login
- Email/mobile verification
- Profile
- Resume upload
- Resume parsing
- Job discovery
- Job details
- Apply
- AJ Interviewer
- Video/audio interview
- AI questions

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- Speech-to-text
- Answer evaluation
- Timer
- Basic tab detection
- Coding environment
- Public/hidden test cases
- Interview report

HR

- HR registration
 - Company profile
 - Create job
 - Configure interview
 - View candidates
 - View AI report
 - Email notification
-

79. Phase 2

Add:

- Advanced integrity monitoring
 - Adaptive interview engine
 - Multiple coding languages
 - Advanced recruiter analytics
 - Skill badges
 - Skill Coins
 - Skill Assistant
 - Blog
 - Courses
 - Personalized learning
 - Notifications
 - Candidate ranking
-

80. Phase 3

Add:

- Stripe subscriptions
 - Premium courses
 - Certificates
 - Advanced AI career coach
 - Advanced assessment engine
 - ATS integrations
 - Enterprise recruiter tools
 - Team hiring
 - Interview libraries
 - Advanced analytics
 - Organization accounts
-

81. Phase 4 — Enterprise

Potential features:

- SSO
 - Advanced RBAC
 - Organization-level analytics
 - Custom AI interview models/prompts
 - Custom scoring systems
 - ATS integrations
 - White-label interviews
 - Enterprise audit logs
 - SLA monitoring
 - Dedicated cloud deployments
-

82. Success Metrics

Candidate

- Application completion rate
- Interview completion rate
- Coding submission rate
- Learning engagement
- Job match rate

HR

- Time saved per candidate
- Time-to-shortlist
- Interview completion
- Assessment accuracy
- Recruiter satisfaction

Platform

Primary KPI:

Qualified Candidate Conversion Rate

Secondary:

- Monthly active candidates
 - Monthly active recruiters
 - Completed interviews
 - Premium conversion
 - Course completion
 - Employer retention
-

83. Functional Acceptance Criteria

Apply

Candidate clicks Apply:

Apply

→ login/registration

→ resume

→ eligibility analysis

→ assessment preparation

→ AJ Interviewer

must work without breaking the original application record.

AI Interview

The interviewer:

- Starts correctly
- Maintains timer
- Asks job-relevant questions
- Adapts follow-up questions
- Records transcript
- Evaluates answers
- Handles reconnection

Coding

Candidate:

- Opens IDE
- Writes code
- Runs code
- Receives test results
- Submits solution
- Gets final coding score

HR

After completion:

Candidate finishes



Analysis completed



HR notified



Candidate appears in HR dashboard

84. Non-Functional Requirements

The system must be:

- Scalable

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- Secure
- Observable
- Mobile-responsive
- Accessible
- Reliable
- Fast
- Maintainable
- Modular

The interview module must degrade gracefully when:

- Microphone permission is denied
 - Camera permission is denied
 - Network temporarily disconnects
 - AI API is temporarily unavailable
 - Speech recognition fails
 - Code execution fails
-

85. Product Differentiation

SkillCortex is not simply another job board.

Its positioning is:

Traditional Job Platform

Resume → Apply → HR → Interview

SkillCortex

Resume



AI Skill Understanding



Job Match



AI Interview

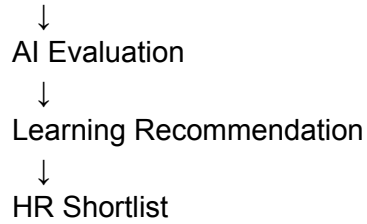


Coding Assessment



Integrity Signals

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This creates a **skills-first recruitment ecosystem**.

86. Brand Positioning

Product Name

SkillCortex

AI Interviewer

AJ Interviewer

AI Career Assistant

Skill Assistant

Learning Currency

Skill Coins

Badges

Skill Badges

Premium

SkillCortex Pro

87. Recommended Visual Design Direction

The UI should communicate:

Intelligent + Human + Technical + Professional

Color system:

Primary:

Black / Deep Charcoal

Surface:

White / Soft Gray

Success / AI:

Light Green

Action:

Modern Blue

Warning:

Amber

Error:

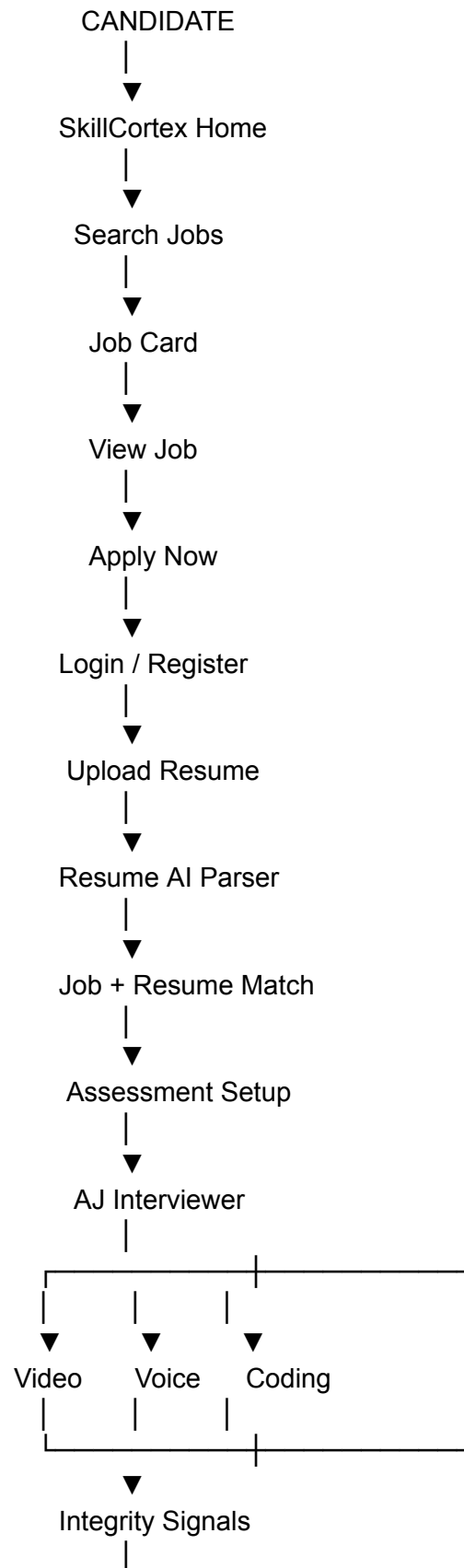
Red

Visual style:

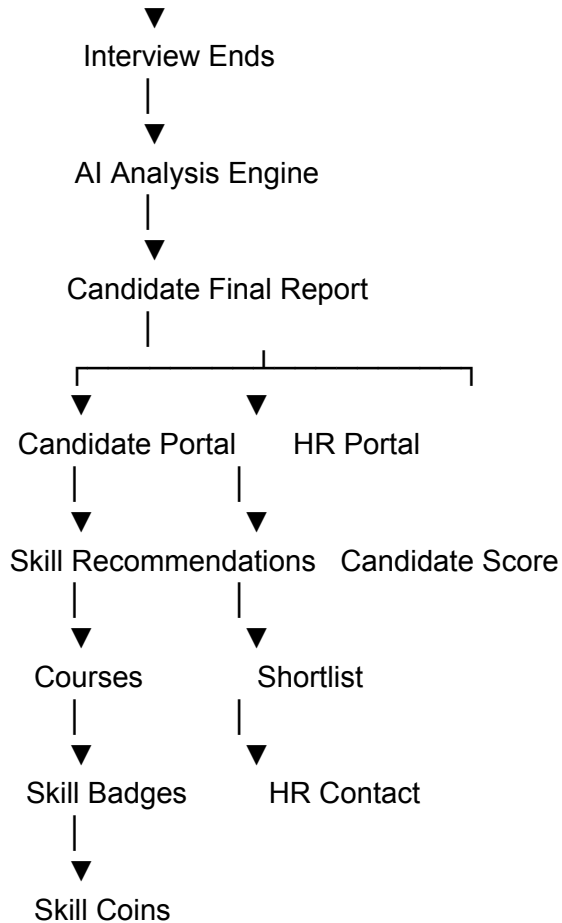
- Large clean cards
- Rounded but not overly playful
- High-quality icons
- Subtle animations
- Minimal shadows
- Glass effects only where meaningful
- Strong typography hierarchy
- Dense information only in recruiter screens
- Spacious candidate experience
- Professional interview workspace

88. Final End-to-End User Journey

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89. Product Principle

The most important product principle for SkillCortex should be:

AI should assist the hiring decision, not become an unexplainable hiring decision-maker.

Every major score should have supporting evidence:

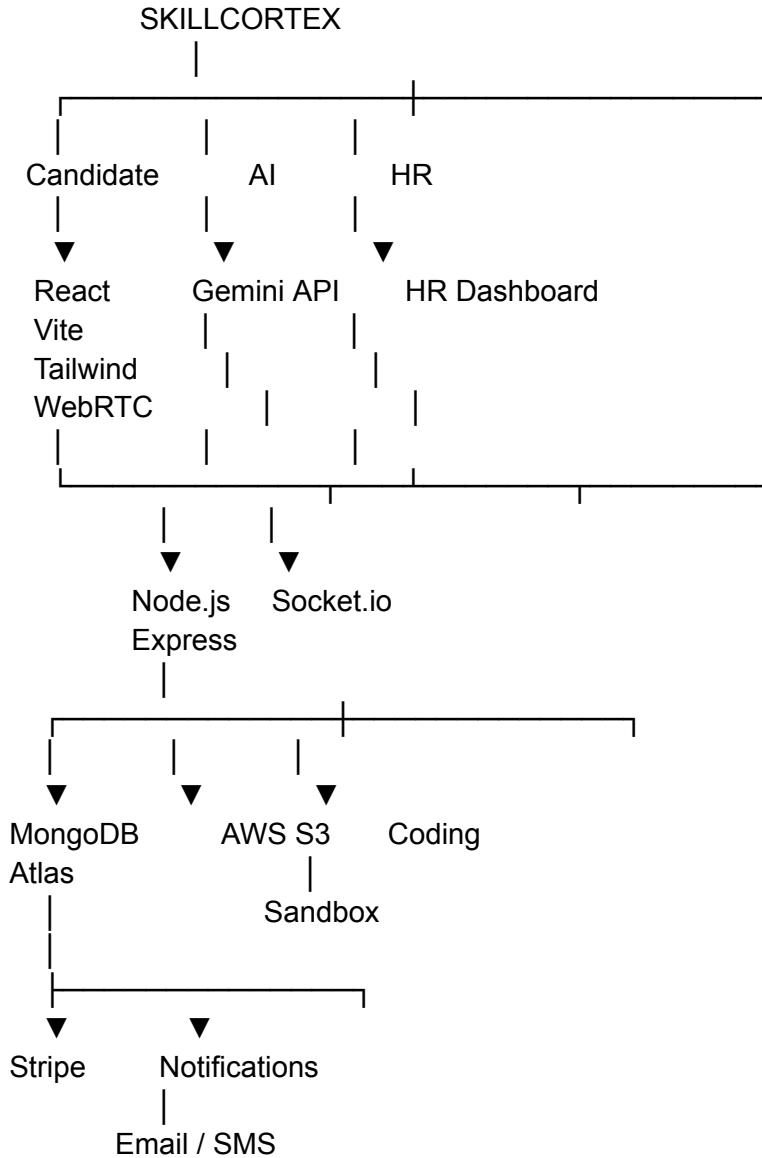
Question → Answer → Evaluation → Evidence → Score

Code → Test Cases → Result → Score

Integrity Event → Timestamp → Evidence → Reviewer Interpretation

This makes the platform more defensible, transparent and useful to recruiters.

90. Final Technical Blueprint



Deployment:

React/Vite → Vercel

Node/Express → AWS EC2

Socket.io → AWS

Files → S3

Database → MongoDB Atlas

Monitoring → CloudWatch

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Payments → Stripe

AI → Gemini

Optional Staging → Render

91. Definition of Done

SkillCortex MVP is considered complete when a real candidate can:

Create account → verify identity → upload resume → browse job → apply → receive personalized AI interview → complete video/voice interview → solve coding challenge → submit → receive AI report → and the corresponding HR receives the candidate assessment automatically.

A recruiter must be able to:

Create company → post job → configure assessment → receive candidate → view resume → view AI interview transcript → review coding results → inspect integrity events → see final score → shortlist/reject candidate.

That is the minimum end-to-end product loop that proves the SkillCortex concept.